香港教育大學
The Education University of Hong Kong

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MTH6184 Data Mining in STEM Education

Chapter 11. Association Rules

Instructor: Philip Yu

Chapter Outcomes

After completing this chapter, you are able to:

- Apply Apriori algorithm to generate association rules (for binary data)
- Evaluate performance of an association rule
- Understand briefly how to extend to non-binary data
- Understand its strengths and weaknesses
- Understand its practical issues
- Final wrap-up

What is Association Rule Mining?

- Association rule mining
  - Given a set of transactions, find association rules that will predict the occurrence of an item based on the occurrences of other items in the transaction
- Examples of association rule mining: retail industry, website traffic analysis and medicine
- List of transactions from a natural foods grocery store:

Transaction number	Items
0	Bread, Milk
1	Bread, Diaper, Beer, Eggs
2	Milk, Diaper, Beer, Coke
3	Bread, Milk, Diaper, Beer
4	Bread, Milk, Diaper, Coke

Example of association rules:

{Diaper} ⇒ {Beer}

{Mike, Bread} ⇒ {Eggs, Coke}

{Beer, Bread} ⇒ {Milk}

Implication means co-occurrence, not causality!

What is Association Rule Mining?

- **Association Rule form:**

  $A \Rightarrow B$: If A then B [support, confidence]

  - Support = P(A and B)
  - Confidence = P(B | A)

- **Example:**
  - Fever = Mild and Cough = Yes $\Rightarrow$ Diagnosis = COVID-19 [40%, 60%]
  - 40% of people have a mild fever and cough, and COVID-19.
  - 60% of people with a mild fever and cough also have COVID-19.

Application: Market Basket Analysis

To mine the sets of items that are frequently bought together at a supermarket by analyzing the customer shopping carts (market baskets).

? Where should detergents be placed in the Store to maximize their sales?

? Are window cleaning products purchased when detergents and orange juice are bought together?

? Is soda typically purchased with bananas? Does the brand of soda make a difference?

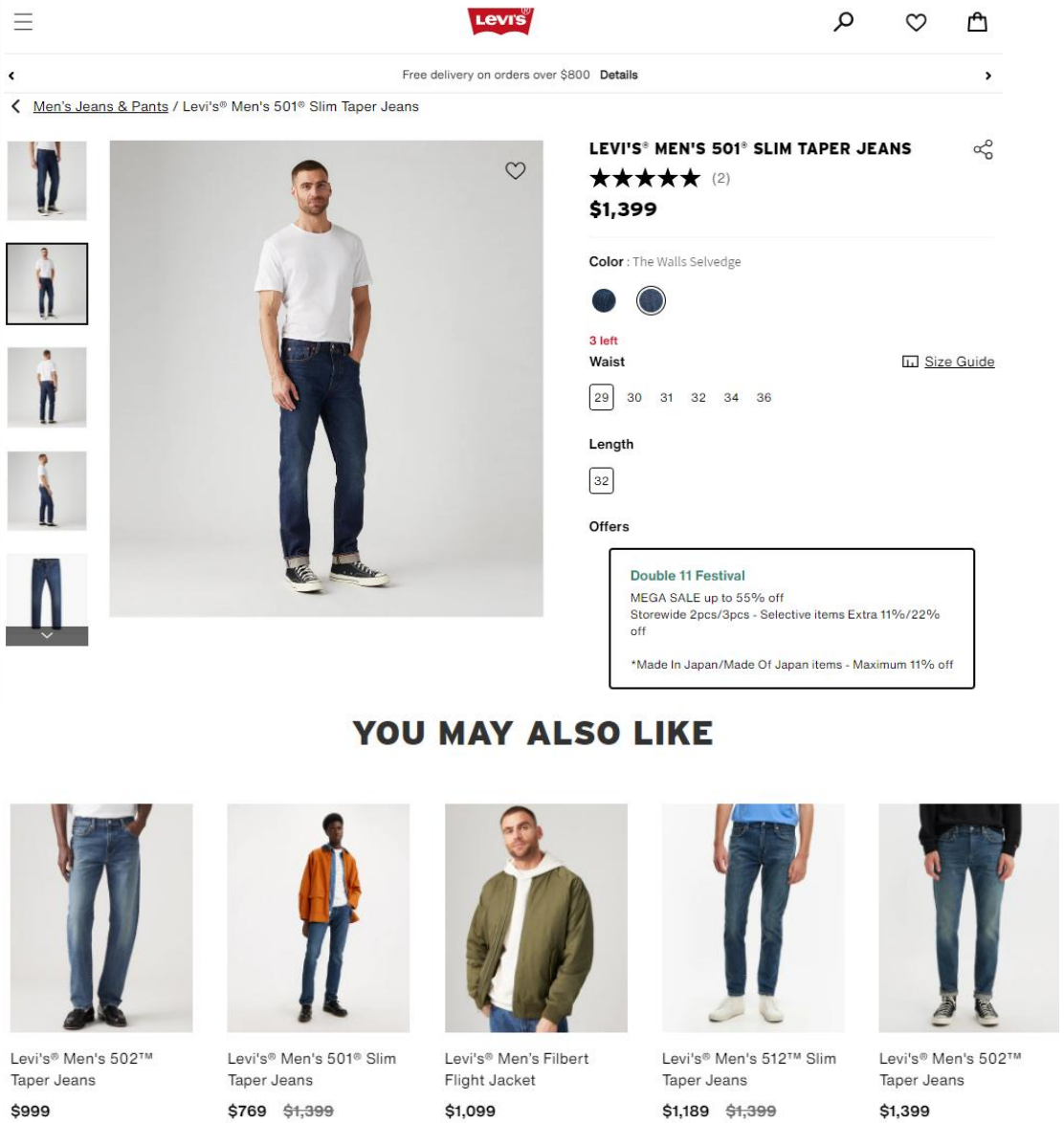
- Goal: increase sales and reduce costs
- Uses:
  - Store layout design
  - Online recommendation engines
  - Cross/up selling
  - Promotion campaign
- Association rule mining is also named as market basket analysis

Bread and Milk $\Rightarrow$ Butter

Famous Example of Market Basket Analysis

- On Friday night, Beer $\Rightarrow$ Diaper
  - A high percentage of beer buyers also buy diapers at the same time. By moving the diapers so that they are within sight of the beer, they can increase the sales of each
  - Possible underlying reason: Probably mom was calling dad at work to buy diapers on way home and he decided to buy a six-pack of beer as well.

Application: Cross-Selling Jeans



- Association rules can be used to encourage cross-selling on a retail website.
- When a male customer looks at one product (Levi's 501 jeans), he is shown other products that might go well with that one.
- The products shown are determined by rules.
- Source: http://www.levi.com/

- Possible Applications:
  - Book / music recommendation
  - Recommendation for Course selection

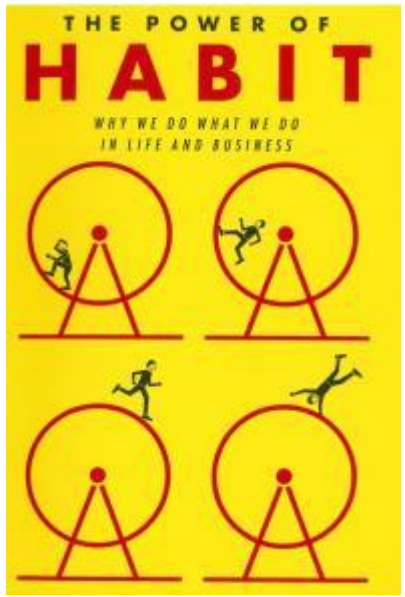
Class Activity 1:

Think what Association Rule was generated

Target knew teen was pregnant before her dad

Target knew teen was pregnant before her dad

- One day, an angry man stormed into a Target store in Minnesota to see a manager.
- “My daughter got this in the mail!” he shouted.
- “She’s still in **high school**, and you’re sending her coupons for **baby clothes and cribs**? Are you trying to encourage her to get **pregnant**?”
- When the manager called the man a few days later to apologize, however, the voice on the other end of the line was conciliatory.
- “I had a talk with my daughter,” he said. “It turns out there’s been some activities in my house I haven’t been completely aware of. **She’s due in August**. I owe you an apology.”



Definition: Frequent Itemset

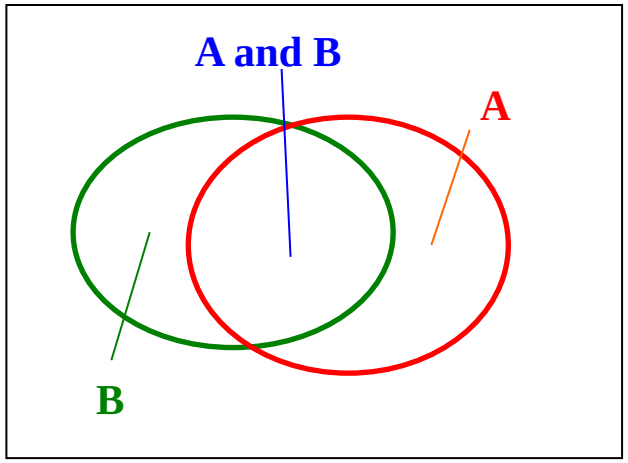
- Itemset
  - A collection of one or more items
    - ✓ Eg: {Milk, Bread, Diaper}
  - k-itemset
    - ✓ An itemset that contains k items
- Support count (σ)
  - Frequency of occurrence of an itemset
  - Eg: σ({Milk, Bread, Diaper}) = 2
- Support (s)
  - Fraction of transactions that contain an itemset
  - Eg: s({Milk, Bread, Diaper}) = 2/5
- Frequent Itemset
  - An itemset whose support is greater than or equal to a *minsup* threshold

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Definition: Association Rule

- Association Rule
  - An implication expression of the form A $\Rightarrow$ B, where A and B are itemsets
  - Example: {Milk, Diaper} $\Rightarrow$ {Beer}
- Rule Evaluation Metrics:
  - Support (s)
    - Fraction of transactions that contain both A and B
    - s(A $\Rightarrow$ B) = P(A and B) = $\frac{\sigma(A \text{ and } B)}{N}$
  - Confidence (c)
    - Measures how often items in B appear in transactions that contain A
    - c(A $\Rightarrow$ B) = P(B | A) = P(A and B)/P(A) = $\frac{\sigma(A \text{ and } B)}{\sigma(A)}$

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke



Example:

$$\{\text{Milk}, \text{Diaper}\} \Rightarrow \text{Beer}$$

$$s = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{\text{N}} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{\sigma(\text{Milk}, \text{Diaper})} = \frac{2}{3} = 0.67$$

Eg 1: Support and Confidence

Rule	Support	Confidence
$A \Rightarrow D$	2/5	2/3
$C \Rightarrow A$	2/5	2/4
$A \Rightarrow C$	2/5	2/3
$B \& C \Rightarrow D$	1/5	1/3

Generating Association Rules using Apriori algorithm

TID	Bread	Milk	Diaper	Beer	Eggs	Coke
1	1	1	0	0	0	0
2	1	0	1	1	1	0
3	0	1	1	1	0	1
4	1	1	1	1	0	0
5	1	1	1	0	0	1

Association Rule Mining Task

- Given a set of transactions, the goal of association rule mining is to find all rules having
  - support ≥ *minsup* threshold
  - confidence ≥ *minconf* threshold
- Many algorithms: eg. Brute-force, Apriori, FP(frequent pattern)-growth, …
- Brute-force algorithm:
  - List all possible association rules
  - Compute the support and confidence for each rule
  - Prune rules that fail the *minsup* and *minconf* thresholds

  ⇒ Computationally prohibitive!
- FP-growth algorithm (Han, et al., 2004) vs Apriori algorithm:
  - FP-growth tends to be faster than Apriori
  - Apriori requires multiple database scan to generate candidate sets but FP-growth requires only two database scans without generating candidate sets

Mining Association Rules

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Rules:

{Milk,Diaper} $\Rightarrow$ {Beer} (s=0.4, c=0.67)
{Milk,Beer} $\Rightarrow$ {Diaper} (s=0.4, c=1.0)
{Diaper,Beer} $\Rightarrow$ {Milk} (s=0.4, c=0.67)
{Beer} $\Rightarrow$ {Milk,Diaper} (s=0.4, c=0.67)
{Diaper} $\Rightarrow$ {Milk,Beer} (s=0.4, c=0.5)
{Milk} $\Rightarrow$ {Diaper,Beer} (s=0.4, c=0.5)

Observations:

- All the above rules are binary partitions of the same itemset:

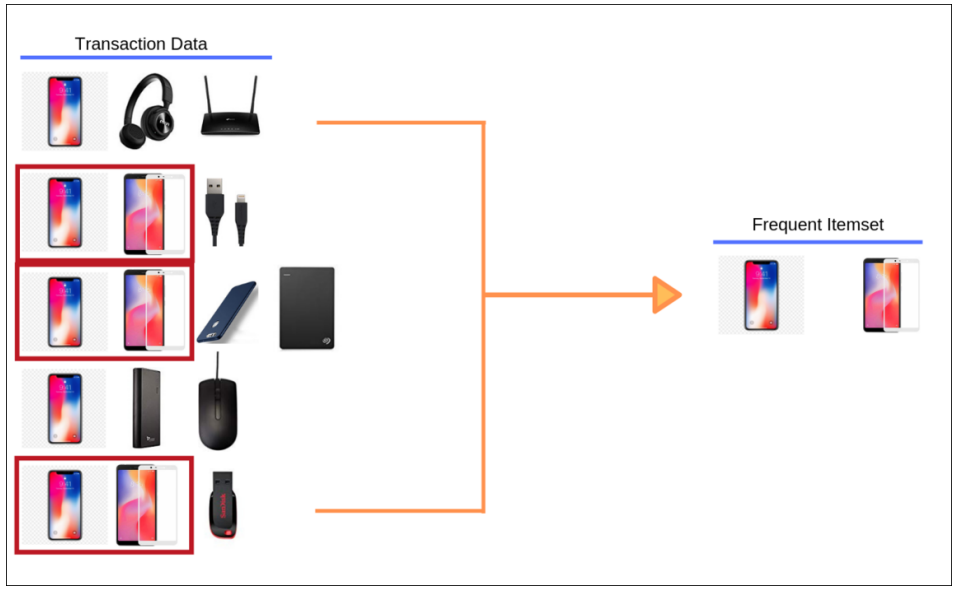
  {Milk, Diaper, Beer}

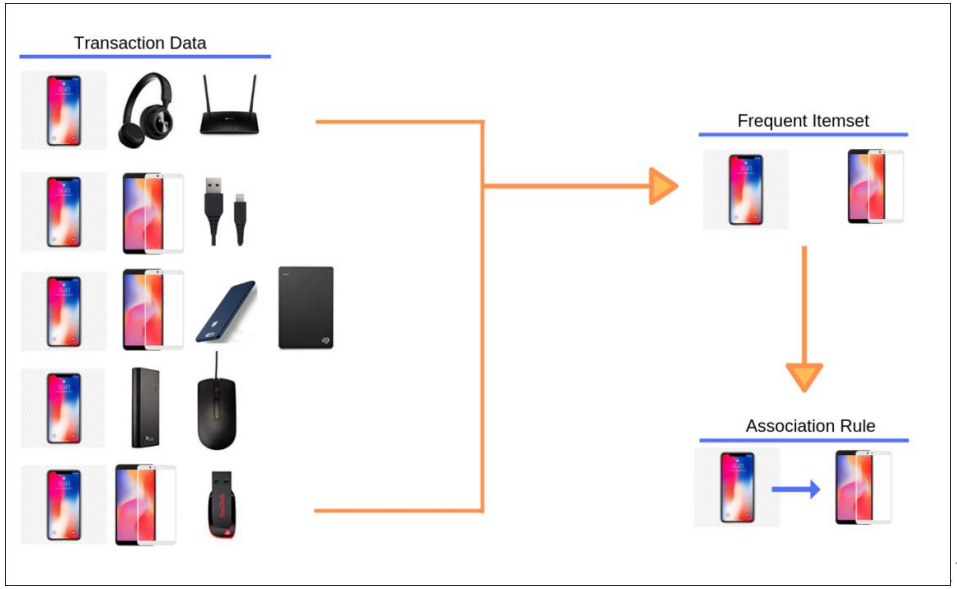
- Rules originating from the same itemset have identical support but can have different confidence
- Thus, we may decouple the support and confidence requirements

Mining Association Rules

```
apriori(transactions,
min_support=0.1, min_confidence = 0.0,
min_lift = 0.0, max_length = None)
```

- Two-step approach:
  1. Frequent Itemset Generation
     - Generate all itemsets whose support ≥ minsup
  2. Rule Generation
     - Generate high confidence rules (≥ minconf) from each frequent itemset, where each rule is a binary partitioning of a frequent itemset
- Both steps are often computationally expensive, particularly the frequent itemset generation.





Frequent Itemset Generation: Apriori Principle

- Frequent (Large) Itemset Property:

  Any subset of a frequent itemset is frequent.

- Contrapositive:

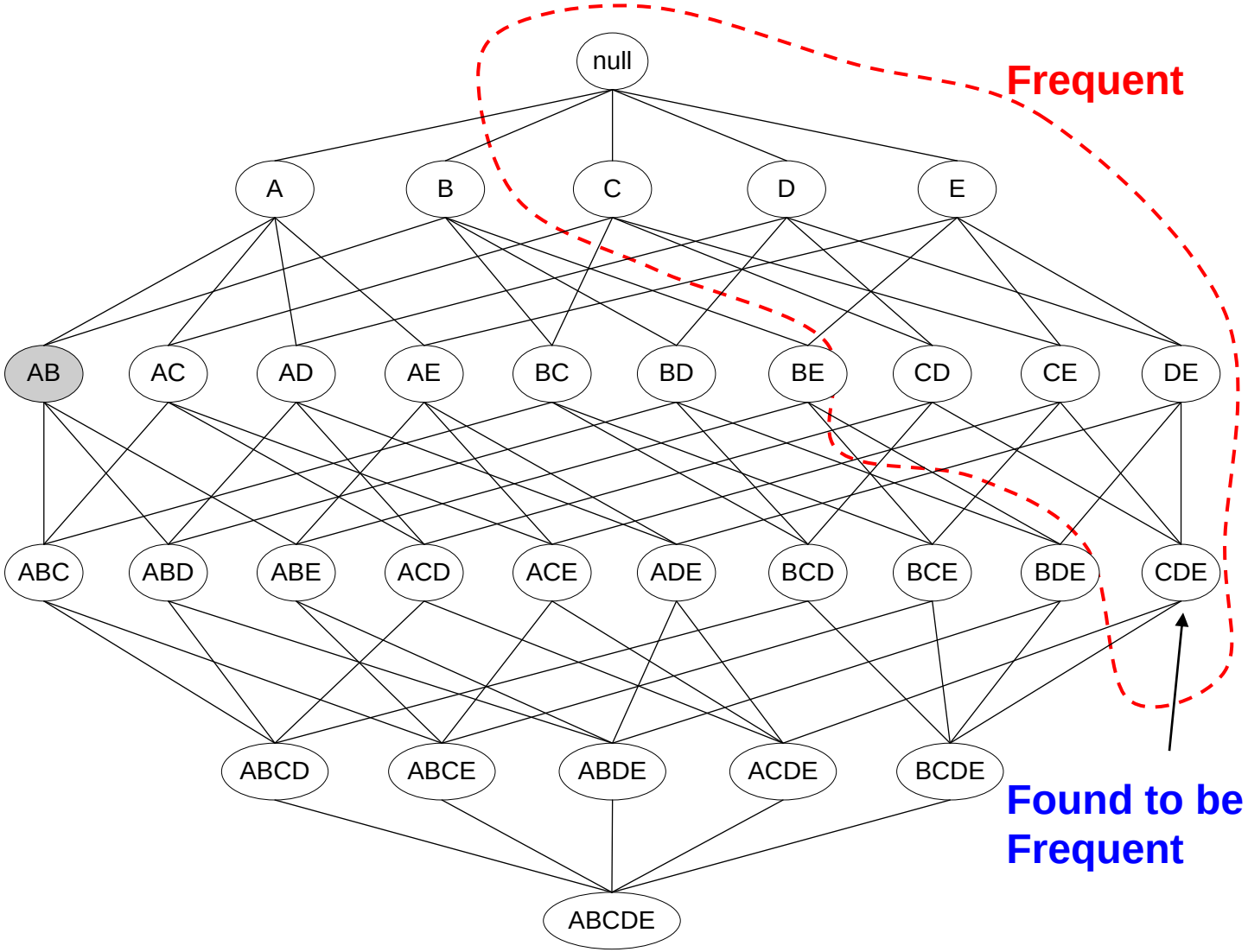
  If an itemset is not frequent,

  none of its supersets are frequent.

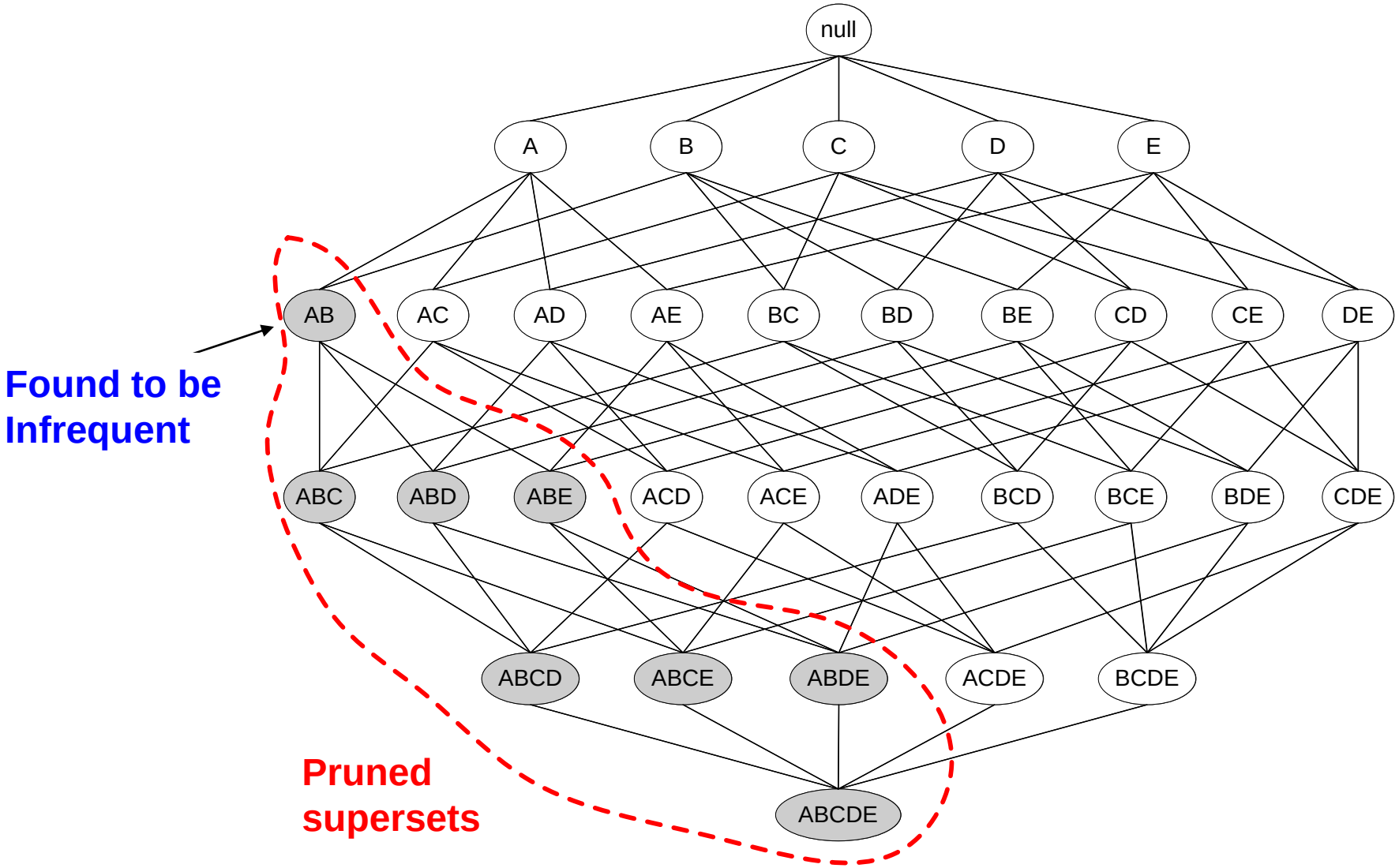
- Rationale: an anti-monotone property

$$\forall A, B : (A \subseteq B) \Rightarrow s(A) \geq s(B)$$

Illustrating Apriori Principle



Illustrating Apriori Principle



Generating Association Rules

- Apriori algorithm: Two steps:
  - Find all frequent itemsets (the sets of items that have minimum support)
    - Find frequent 1-itemsets, L_1
    - Use L_1 to find frequent 2-itemsets, L_2
    - Use L_2 to find frequent 3-itemsets, L_3, and so on
    - Until no more frequent k-itemsets can be found
  - Generate rules from the frequent itemsets
    - rules that have minimum support and minimum confidence (discuss later)

Eg 2: Apriori Algorithm with minimum support = 2

Database D

TID	Items
100	1 3 4
200	2 3 5
300	1 2 3 5
400	2 5

Scan D →

C_1

itemset	sup.
{1}	2
{2}	3
{3}	3
{4}	1
{5}	3

→

L_1

itemset	sup.
{1}	2
{2}	3
{3}	3
{5}	3

C_2

itemset
{1 2}
{1 3}
{1 5}
{2 3}
{2 5}
{3 5}

Scan D ←

C_2

itemset	sup
{1 2}	1
{1 3}	2
{1 5}	1
{2 3}	2
{2 5}	3
{3 5}	2

←

L_2

itemset	sup
{1 3}	2
{2 3}	2
{2 5}	3
{3 5}	2

C_3

itemset
{2 3 5}

Scan D →

L_3

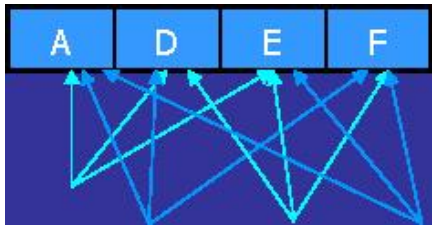
itemset	sup
{2 3 5}	2

How to generate C_k from L_{k-1}

- Many methods: eg. Brute-force method
- Note the Apriori Principle implies that

  "a k-itemset is generated only if all subsets of size k-1 are frequent (k-1)-itemsets"
- Step 1: Self-joining (L_{k-1} x L_{k-1})
  - Suppose the items in L_{k-1} are listed in an order
  - Candidate rule is generated by merging a pair of frequent (k-1)-itemsets in L_{k-1} only if their first k-2 items are identical.
  - Eg. {1,3,4,5} x {1,3,4,6} gives {1,3,4,5,6}

- Step 2: Pruning
  - The k-itemset found in Step 1 is not frequent if any one of its subset of size k-1 is not in L_{k-1} (i.e. not frequent)



Eg 3: Generate C_4 from L_3

- L_3 = {{a,b,c},{a,b,d},{a,c,d},{a,c,e},{b,c,d}}
- Join step:
  - $C_4 = L_3 \times L_3$ = {{a,b,c,d}, {a,c,d,e}}
- Prune step:
  - {a,b,c,d}:
    - check if {a,b,c},{a,b,d},{a,c,d},{b,c,d} are in L_3
  - {a,c,d,e}:
    - check if {a,c,d},{a,c,e},{a,d,e},{c,d,e} are in L_3
- Thus C_4 = {{a,b,c,d}}

Eg 4: A Woman's clothing store: transactions

Trans-ID	Items
T1	Blouse
T2	Shoes, Skirt, TShirt
T3	Jeans, TShirt
T4	Jeans, Shoes, TShirt
T5	Jeans, Shorts
T6	Shoes, TShirt
T7	Jeans, Skirt
T8	Jeans, Shoes, Shorts, TShirt
T9	Jeans
T10	Jeans, Shoes, TShirt
T11	TShirt
T12	Blouse, Jeans, Shoes, Skirt, TShirt
T13	Jeans, Shoes, Shorts, TShirt
T14	Shoes, Skirt, TShirt
T15	Jeans, TShirt
T16	Skirt, TShirt
T17	Blouse, Jeans, Skirt
T18	Jeans, Shoes, Shorts, TShirt
T19	Jeans
T20	Jeans, Shoes, Shorts, TShirt

Result: Candidates (C) and Frequent Itemsets (L)

Scan	Candidates (Support)	Frequent Itemsets (min. sup=4)
1	{Blouse}(3), {Jeans}(14), {Shoes}(10), {Shorts}(5), {Skirt}(6), {TShirt}(14)	{Jeans}, {Shoes}, {Shorts}, {Skirt}, {Tshirt}
2	{Jeans, Shoes}(7), {Jeans, Shorts}(5), {Jeans, Skirt}(3), {Jeans, TShirt}(9), {Shoes, Shorts}(4), {Shoes, Skirt}(3), {Shoes, TShirt}(10), {Shorts, Skirt}(0), {Shorts, TShirt}(4), {Skirt, TShirt}(4)	{Jeans, Shoes}, {Jeans, Shorts}, {Jeans, TShirt}, {Shoes, Shorts}, {Shoes, TShirt}, {Shorts, TShirt}, {Skirt, TShirt}
3	{Jeans, Shoes, Shorts}(4), {Jeans, Shoes, TShirt}(7), {Jeans, Shorts, TShirt}(4), {Shoes, Shorts, TShirt}(4)	{Jeans, Shoes, Shorts}, {Jeans, Shoes, TShirt}, {Jeans, Shorts, TShirt}, {Shoes, Shorts, Tshirt}
4	{Jeans, Shoes, Shorts, Tshirt}(4)	{Jeans, Shoes, Shorts, TShirt}
5	Empty	Empty

Class Activity 2:
Self-joining and Pruning

Rule Generation

- Given a frequent itemset L, find all non-empty subsets $f \subset L$ such that $f \Rightarrow L - f$ satisfies the minimum confidence requirement
  - If {A,B,C,D} is a frequent itemset, candidate rules:

    $ABC \Rightarrow D$, $ABD \Rightarrow C$, $ACD \Rightarrow B$, $BCD \Rightarrow A$,

    $AB \Rightarrow CD$, $AC \Rightarrow BD$, $AD \Rightarrow BC$, $BC \Rightarrow AD$, $BD \Rightarrow AC$, $CD \Rightarrow AB$,

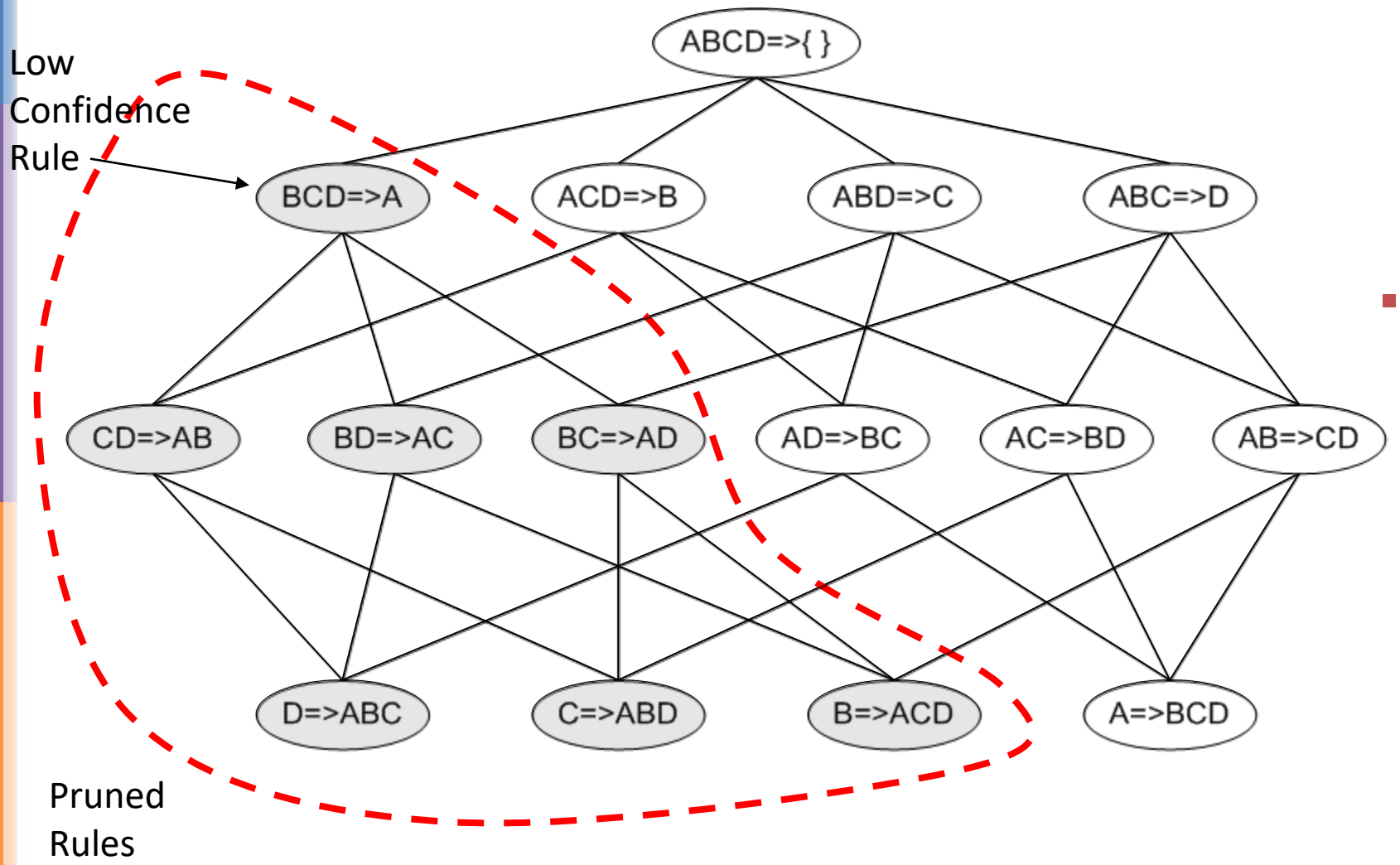
    $A \Rightarrow BCD$, $B \Rightarrow ACD$, $C \Rightarrow ABD$, $D \Rightarrow ABC$
- If $|L| = k$, then there are $2^k - 2$ candidate association rules (ignoring $L \Rightarrow \varnothing$ and $\varnothing \Rightarrow L$).
- Note that all these $2^k - 2$ rules have already met the minimum support requirement (Why?)
- How to efficiently generate rules from frequent itemsets?
  - In general, confidence does not have an anti-monotone property

    $c(ABC \Rightarrow D)$ can be larger or smaller than $c(AB \Rightarrow D)$
  - But confidence of rules generated from the *same itemset* has an anti-monotone property
  - e.g., L = {A,B,C,D}:

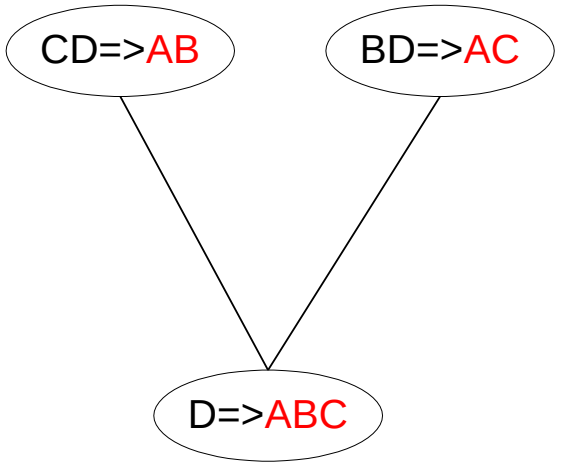
    $$c(ABC \Rightarrow D) \geq c(AB \Rightarrow CD) \geq c(A \Rightarrow BCD)$$
  - Confidence is anti-monotone w.r.t. number of items on the RHS of the rule

Rule Generation in Apriori Algorithm

Lattice of rules



- Similar to L_{k-1} x L_{k-1} method, a candidate rule of k consequents can be generated by merging two rules with identical first k-2 consequents
- Merge (CD=>AB, BD=>AC) would produce the candidate rule D => ABC



Eg 5: Generating Rules

Trans-ID	Items
1	ACD
2	BCE
3	ABCE
4	BE
5	ABCE

Min support = 60%
Min confidence = 75%

Frequent itemset	Support
BCE	60%
AC	60%
BC	60%
CE	60%
BE	80%
A	60%
B	80%
C	80%
E	80%

Rule	Confidence
{BC} => {E}	100%
{BE} => {C}	75%
{CE} => {B}	100%
{B} => {CE}	75%
{C} => {BE}	75%
{E} => {BC}	75%

Evaluation of Association Rules

X	1	1	1	1	0	0	0	0
Y	1	1	0	0	0	0	0	0
Z	0	1	1	1	1	1	1	1

Pattern Evaluation

- Association rule algorithms can produce large number of rules
  - many of them are uninteresting or redundant
  - Eg. Redundant if {A,B,C} ⇒ {D} and {A,B} ⇒ {D} have same support & confidence
- How can we measure the quality / strength of a rule?
  - So far, support & confidence are the only measures used
- **Example:** Saving & checking accounts

		Checking Account		
		No	Yes	
Saving Account	No	500	3,500	4,000
	Yes	1,000	5,000	6,000
				10,000

Support(SVG ⇒ CK) = 50%

Confidence(SVG ⇒ CK) = 83%

Is this rule useful?

Criticism to Support & Confidence

- **Example:** (Aggarwal & Yu, PODS98)
  - Among 5000 students
    - 3000 play basketball
    - 3750 eat cereal
    - 2000 both play basket ball and eat cereal

	Basketball	Not basketball	Sum (row)
Cereal	2000	1750	3750
Not cereal	1000	250	1250
Sum(col)	3000	2000	5000

  - *play basketball* $\Rightarrow$ *eat cereal* [40%, 66.7%] is misleading because the overall percentage of students eating cereal is 75% which is higher than 66.7%.
  - *play basketball* $\Rightarrow$ *not eat cereal* [20%, 33.3%] is far more informative, although with lower support and confidence

Criticism to Support & Confidence

- Example:
  - X and Y: positively correlated,
  - X and Z, negatively correlated
  - support and confidence of X ⇒ Z dominates

X	1	1	1	1	0	0	0	0
Y	1	1	0	0	0	0	0	0
Z	0	1	1	1	1	1	1	1

- We need a dependency or correlation measure between two events

$$Corr(X,Y) = \frac{P(X \text{ and } Y)}{P(X)P(Y)}$$

- This correlation is also called the lift of rule X ⇒ Y (or Y ⇒ X)

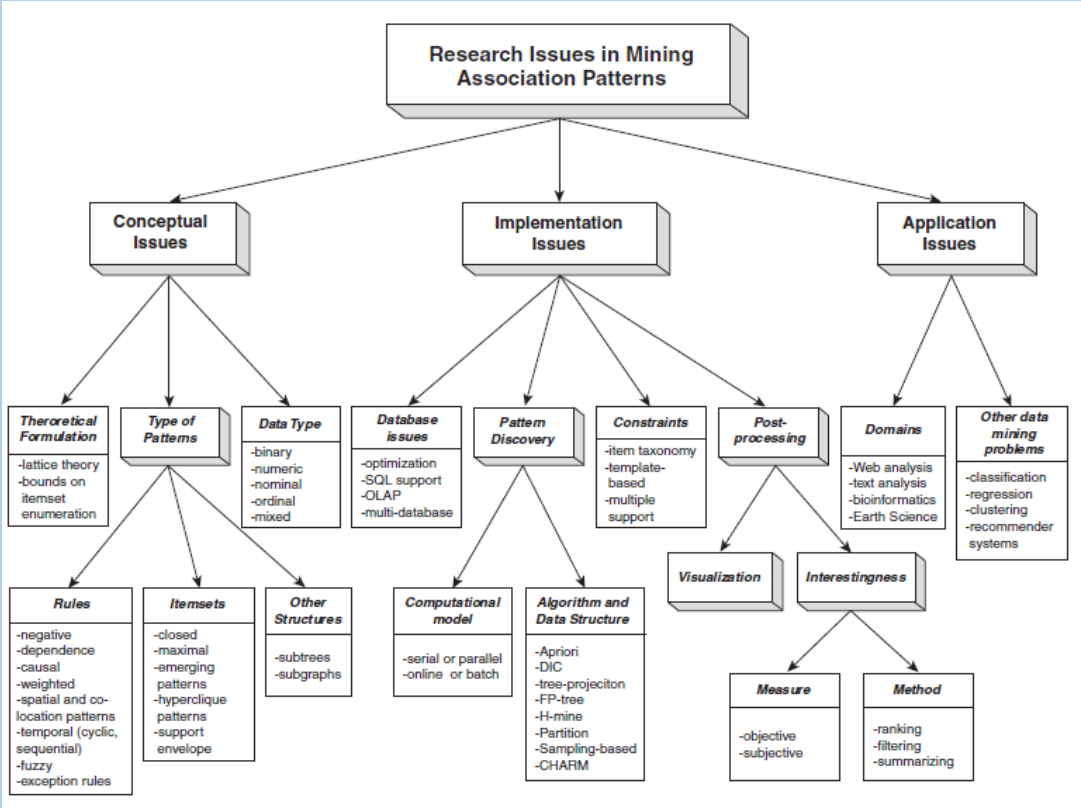
$$\text{Lift of } (X \to Y) = \frac{P(X \text{ and } Y)}{P(X)P(Y)} = \frac{\text{confidence}(X \to Y)}{P(Y)}$$

- Lift compares the confidence of the rule with the expected confidence when A and B are independent.
- P(A and B) = P(A)P(B), if A and B are independent events
- A and B negatively correlated, if the lift value is less than 1; otherwise A and B positively correlated
- Lift > 1 indicates that the rule is useful

Rule	Support	Confidence
$X \to Y$	25%	50%
$X \to Z$	37.5%	75%

Itemset	Support	Lift
X, Y	25%	2
X, Z	37.5%	0.9
Y, Z	12.5%	0.57

Extension of Association Rule Mining



Extension of Association Rule Mining

- Association vs Dissociation rules
- Binary vs. Continuous / Categorical attributes

- **Examples:**
  - buy “PC” and buy “printer” $\Rightarrow$ buy “software”
  - don’t buy “DVD drive” $\Rightarrow$ buy “CDRW drive”
  - age “30..39” and income “42..48K” $\Rightarrow$ buy “PC”
  - age “54..64” and home “Amoy” $\Rightarrow$ status “death”

Handling Dissociation rules and Continuous / Categorical Inputs

Gender	···	Age	Annual Income	No of hours spent online per week	No of email accounts	Privacy Concern
Female	···	26	90K	20	4	Yes
Male	···	51	135K	10	2	No
Male	···	29	80K	10	3	Yes
Female	···	45	120K	15	3	Yes
Female	···	31	95K	20	5	Yes
Male	···	25	55K	25	5	Yes
Male	···	37	100K	10	1	No
Male	···	41	65K	8	2	No
Female	···	26	85K	12	1	No
···	···	···	···	···	···	···

Male	Female	··· ···	Age < 13	Age $\in [13, 21)$	Age $\in [21, 30)$	··· ···	Privacy = Yes	Privacy = No
0	1	···	0	0	1	···	1	0
1	0	···	0	0	0	···	0	1
1	0	···	0	0	1	···	1	0
0	1	···	0	0	0	···	1	0
0	1	···	0	0	0	···	1	0
1	0	···	0	0	1	···	1	0
1	0	···	0	0	0	···	0	1
1	0	···	0	0	0	···	0	1
0	1	···	0	0	1	···	0	1
···	···	···	···	···	···	···	···	···

Class Activity 3:
Association Mining for Heart Disease Diagnosis

Pros and Cons of Association Rule Mining

- Pros
  - It produces easy to understand results
  - It works on mixed type data
  - Rules are relatively easy to compute
- Cons
  - It is an exponentially growth algorithm
  - It usually create many rules and some rules are not useful
  - It discounts rare items

Practical Issues

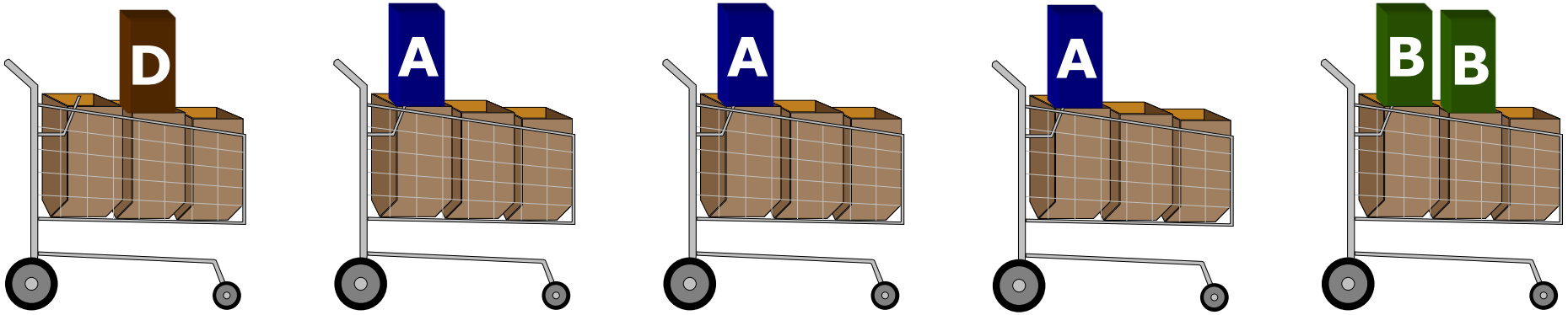
- Practical problems often have sparse data: many input attributes, few items per transaction
- Goal is typically search for high support
  - High support = broad impact
    - How high support is high? 70%? 20%?
    - Start with a high support and lower it
    - If support of a large itemset is close to 100%, then remove this itemset and repeat association rule mining
  - High confidence not very crucial (as opposed to classification)
    - Eg. play basketball $\Rightarrow$ not eat cereal [20%, 33.3%]
    - Confidence around 50%-70% may be sufficient
    - We might only use lift to select useful association rules
- Users can search for useful information by taking a subset of the rules:
  - Only find rules involving the dairy department
  - Only find rules involving expensive products
  - Only find "expensive" rules
  - Only find rules with "whiskey" on the right hand side
  - Only find rules with "milk" on the left hand side
  - Calendars (every Sunday, every 1st of the month)

Practical Issues

- When is Association Rule Useful?
  - The useful rule contains high quality, actionable information.
  - Trivial results are already known by anyone at all familiar with the business.
  - Inexplicable results seem to have no explanation and do not suggest a course of action.
  - Trivial and inexplicable rules occur most often.

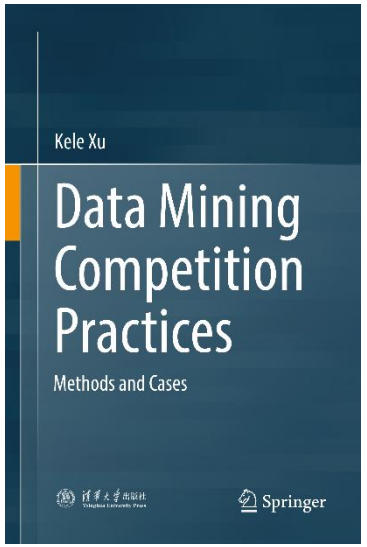
- Useful: On Fridays convenience store customers often purchase diapers and beer together.
- Trivial: Customers who purchase maintenance agreements are very likely to purchase large appliances.
- Inexplicable: When a new Super Store opens, one of the most commonly sold item is light bulbs.

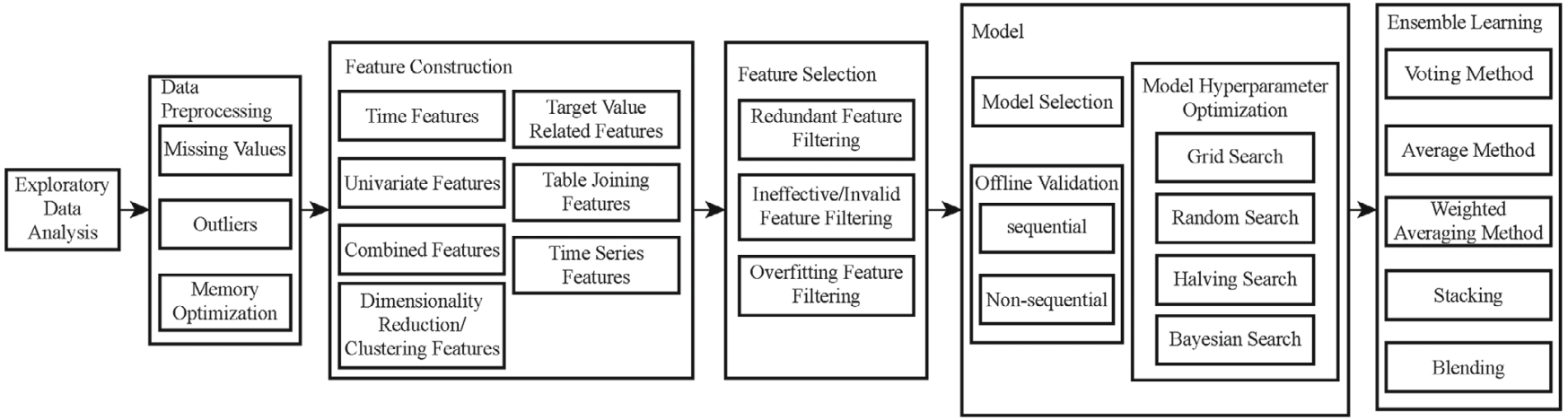
- If very few transactions involved multiple items, will MBA work?



Final Wrap-Up

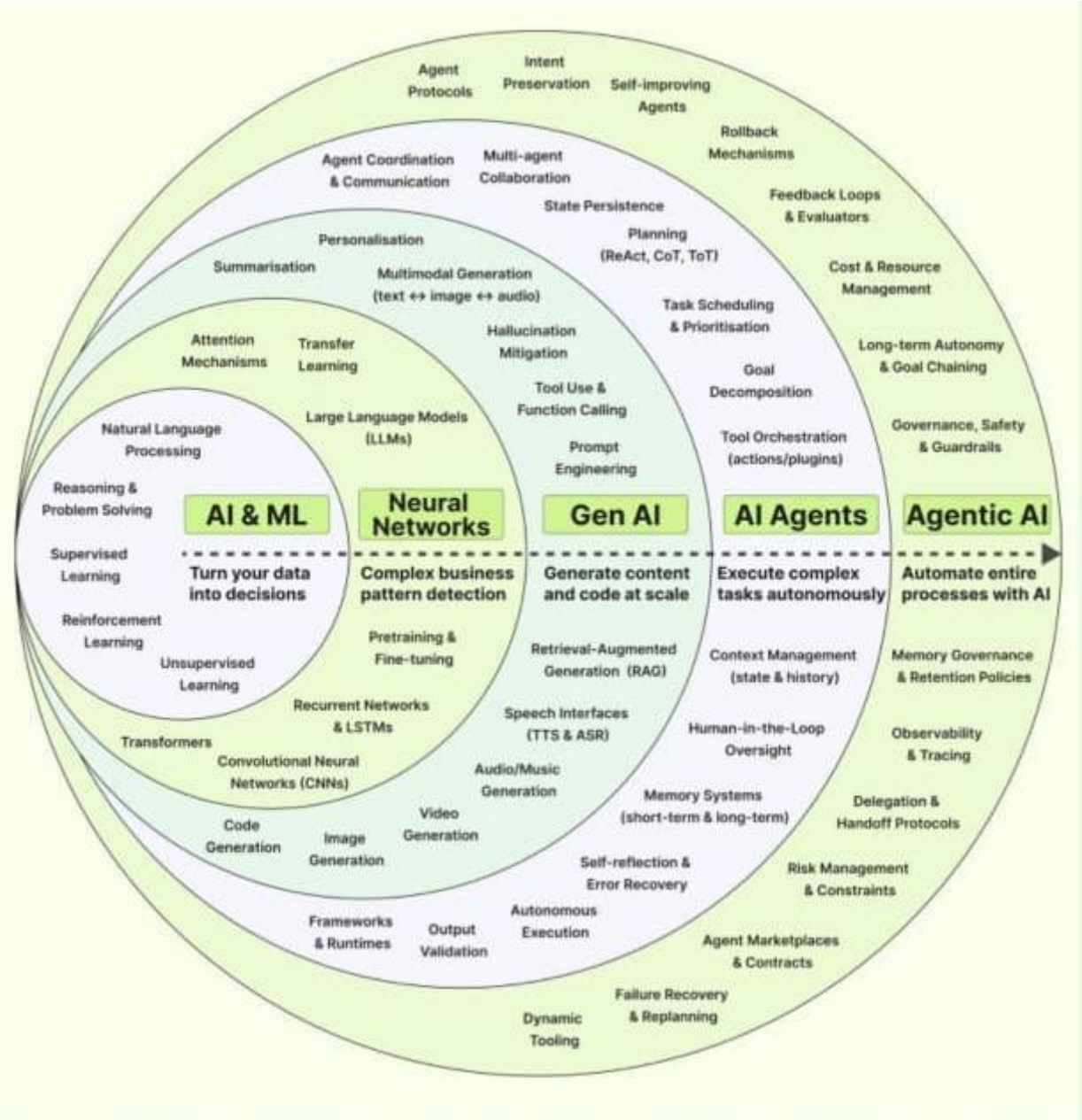
- Covered major data mining methods and STEM designs:
  - Regression, Decision Tree, kNN, Naïve Bayes, Ensemble Methods, ANN, Text Mining, Clustering, Association Rules
- Course Learning Outcomes:
  1. describe key elements of mathematical theories for data mining, data mining processes and algorithms;
  2. demonstrate an understanding of applying STEM education at schools;
  3. select appropriate methods to represent and analyse different data for different tasks in STEM education;
  4. process and analyse data to make informed decisions with statistical and data mining principles; and
  5. reflect critically on the importance of STEM to address educational problems and the awareness of STEM professions.





The Future of AI

- AI & ML: Turn your data into decisions
- Neural Networks: Handle complex pattern detection
- GenAI: Generate content and code at scale (eg. Write a story)
- AI Agents: Execute complex tasks autonomously (eg. Book a flight)
- Agentic AI: Automate entire processes with AI through long-term autonomy and multi-agent collaboration (eg. Plan a full holiday (visa, flights, hotels))



THANK YOU